Loki III: Reverse Engineering Structural Netlists without a Language Model
How far do provenance, a Bayesian L1 and a graph L2 get on their own?

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***Abstract*—Gate-Breaker showed that a language model can recover functional RTL from a gate-level netlist, and it ended by naming the graph techniques its model needed: clustering to name buses, a classification of signals as datapath or control, and partitioning. Loki III asks the complementary question: how much structure can be recovered *without* a language model, from provenance, a Bayesian hypothesis flow and graph analysis alone? This paper describes the benchmarks built to answer that question and reports the recovery statistics under the Gate-Breaker metrics. The benchmarks are 68 synthesizable micro designs and the 156 VerilogEval references (150 build on three floors), compiled by a provenance-tracked front end into named, labelled, anonymised netlist datasets; a renamed-instance protocol then proves the pipeline naming-neutral across 105,646 renamed identifiers. On the L1 hypothesis flow, the structures that are already reverse engineered by the synthesis flow recover almost perfectly (hard adders 67 of 67, hard multipliers 30 of 30), while the structures the flow erased recover weakly (macro-F1 0.354 on the first floor, 0.108 on the second, calibrated). A second round of compute-only experiments finds that re-mapping the gate netlist to LUT6 brings its in-sample recovery to the second floor's level (not on the held-out designs, where the result is inconclusive), that the language model is more readable than Loki on the same designs while Loki's output is formally equal to its input, and that an operator-recovery F1 sees what AST and CFG similarity cannot. The L2 graph stage beats a 1-WL baseline on real micros once node, graphlet, WL and NPN features combine (macro-F1 0.4241 against 0.2717) and its clusters carry the designer's hierarchy (Louvain modularity 0.4220). We state the honest limits: no language model step exists, the recovered structure is structural rather than functional, and the paper's own determinism defects are reported.**

***Index Terms*—reverse engineering, netlist, provenance, Bayesian inference, graph neural networks, FPGA CAD**

I. Introduction

Gate-Breaker [1] takes a gate-level netlist with obfuscated wire names, formats it into a four-step prompt (interface, decomposition, function, generation), asks a language model to write RTL, and checks the result by simulation. On 156 VerilogEval problems [2] its best model recovered functional RTL for 75.1% of the runs, and 53.4% on the hardest quartile. The paper's own conclusion is the interesting part: external graph tools should help the model – clustering to name buses, a classification of signals as datapath or control, and partitioning of the netlist into smaller parts.

Loki III (https://github.com/drpaj12/loki) is the Yggdrasil III suite's reverse-engineering tool, and it targets exactly those three items – with no language model step at all. Its pipeline is L0, a provenance ground-truth dataset made from the benchmarks; L1, a Bayesian hypothesis flow whose pattern detectors propose structures and whose proofs confirm them; and L2, graph analysis (clustering, structural features, graph neural networks) over the same netlists. The research question this paper answers is: *how good are the L0, L1 and L2 experiments, without more intelligence, at recovering structure?* The answer matters for two audiences. For the Gate-Breaker line, it prices the graph side of the future work the paper names. For reverse engineering, it measures what a provenance-tracked front end and honest statistical machinery can do before any model is added.

The contributions are: (1) a benchmark method – provenance-tracked, labelled, anonymised netlist datasets with a renamed-instance protocol that proves naming neutrality, so the pipeline can be evaluated on designs whose names carry nothing; (2) the L1 and L2 recovery statistics under the Gate-Breaker metrics and under Loki's own (precision, recall, calibration, clustering quality, GNN against a 1-WL baseline); (3) the honest limits, including two determinism defects the experiments found.

II. Background

A. The provenance problem and Odin III

Synthesis modifies the designer's functional intention: flattening removes hierarchy, optimization merges operators, and mapping covers logic with cells that carry no source names. Linking each netlist object back to the HDL that produced it is the provenance problem [3]. Odin III, the provenance-tracked front end under Loki, keeps that link: every object of its one IR records its source location, and the link survives elaboration, optimization, partial mapping and the linked ABC flow. Loki stands on that link: its L0 datasets carry the answer key the

synthesis tool kept, which is what makes the recovery statistics in this paper possible at all.

B. Gate-Breaker as the method baseline

Gate-Breaker measures recovery by functional correctness and by the similarity of the recovered and reference RTL: Success Rate (the fraction of K samples that simulate correctly), Success@k (the fraction of designs solved within the first k samples), AST similarity (mean subtree Jaccard over normalized identifiers, depth $d = 3$) and CFG similarity (a normalized shortest-path node kernel over control-flow graphs). Its complexity score weights sequential blocks, generates, loops, instantiations, always blocks, buses, multiplexers, cases, arithmetic operators, assignments and concatenations into quartiles Q1 (simple combinational) through Q4 (complex sequential). Loki reports its own statistics on the same axes where they apply – recovery rate per structure kind, calibration error, clustering quality against the known hierarchy, and GNN accuracy against a 1-WL baseline – and the Gate-Breaker metrics on the replication runs.

III. Methodology: floors, pipeline and evaluation

A. Floors: the abstraction level of the netlist

A *floor* is the abstraction level and technology mapping at which an Odin III design is cut and written out as the L0 netlist, the object the reconstruction receives. Every floor starts from the same elaborated design, the common front `proc; fsm; opt; flatten; opt -full; wreduce` (`FRONT` in `loki/odin.py`), and then maps it with a floor-specific script (`floor_script` in the same file). Table I lists the floors. Four exist: `f1-k6` and `f1-earch` (one floor per FPGA architecture file; the directory name is `f1-` plus the architecture's lower-case prefix), `f2`, and `f3`.

What a floor's netlist contains decides what is left to reconstruct. At `f1` the flow's `hardmap` step has already recognised adders, multipliers and memories, so they appear as typed hard cells; the construct they implement is part of the netlist, and the labels still name it. At `f2` the same logic is only LUT truth tables: an adder is a carry chain of LUTs that has to be found again. At `f3` the logic is a network of two-input gates, the form Gate-Breaker's language model receives. The `f2` output does not depend on the architecture file, and `f3` reads none, so their files are byte-identical across architectures.

Loki uses several floors because what a synthesis flow erases depends on the mapping, and recovery has to be measured against each. `f1` keeps the structures the flow already reverse engineered, which is what lets the `hard` detector of L1 read them from Odin III's `unmap` facts; `f2` removes that crutch with the same source and the same answer key; `f3` makes the input comparable to Gate-Breaker's and is the floor on which the AST and CFG similarities are computed (Section V-C). Within one design the floors share one source of truth, the provenance spans of the same elaborated IR, so a difference between floors is a difference of the mapping.

B. L0: provenance ground-truth datasets

The first benchmark set is the 68-design micro suite (counters, registers, arithmetic, ALUs, muxes, comparators, small FSMs) that Odin III already compiles and checks against Yosys; the second, the 156 VerilogEval references, is described in Section IV. L0 compiles each design on every floor of Table I. Every netlist object carries its provenance (source file and line), a neutral `n`k name from one shared counter, and a label: the structure kind (`construct_type`) is the answer key in `labels.csv`, and the netlist-visible hard-block kind (`cell_type`) travels in the graph labels, so structures that are already reverse engineered stay visible to the reconstruction. `loki check` validates every dataset: 68 designs, 269 floors, a label row for every node, zero errors. The two hard-block designs the v1.0 front end refused now compile on the v1.1.0 SystemVerilog front end, so the set is complete.

Renamed instances. A separate protocol asks the naming question directly: rename every identifier of every netlist (105,646 identifiers across 269 netlists) through a shared permutation, run the pipeline on the renamed floors, and inverse-map every token back. The verifier reports token-for-token equality with the original, zero failures, and the base and renamed score digests are byte-identical (sha256 over all run scores, 204 of 204 pairs) – the pipeline is naming-neutral, so nothing it recovers is an artefact of a name.

C. L1: the Bayesian hypothesis flow

L1 is a round-based hypothesis flow over one netlist, with no language model. It reads a floor netlist into a fresh Odin III design (level 0, never edited) in which every cell is a function, and keeps a working copy that grows one level of hierarchy per committing round.

Hypotheses and detectors. A hypothesis is a claim that a set of level-0 cells is a structure of some kind, for example: a flip-flop with enable or synchronous reset (`ff_en`, `ff_srst`), a `register`, a `bus` of isomorphic slices, an `adder`, `subtractor`, `comparator` or constant comparison (`eq_const`), a `mux` or `mux_tree`, a `counter`, `shift_reg`, `lfsr` or `fsm`, a datapath or control `unit`, a per-cell `role`, or a hard block (`hard_add`, `hard_mul`, `hard_mem`). Nine detector plugins (arith, bus, ff_pattern, fsm, hard, mux, role, seqfn, unit) propose them. A detector reports features as *bin labels* (a width of 5 to 8, whether an enable function is exact, the NPN class of a cell) and never a likelihood; the likelihoods come from calibration.

Bayesian inference. Each hypothesis h of kind k from detector d carries a log-odds $L(h)$, the logarithm of the odds that it is a real structure, and $p(h) = 1/(1 + e^{-L(h)})$ is its probability. L is a naive-Bayes sum: a prior plus evidence items, each the log-likelihood ratio (LLR) of an observation,

$$L(h) = \pi_{d,k} + \mathrm{clip}_{\pm 6.91}\Big(\sum_f w_{\phi(f)}\, \lambda_{k,f}(b_f) + \min\big(1.10\, n_{\mathrm{conf}},\, 3.30\big) - 2.30\, n_{\mathrm{ref}} \Big) + e_h, \quad (1)$$

TABLE I
THE FLOORS OF L0. ALL FLOORS RUN THE COMMON FRONT FIRST; THE MAPPING SCRIPT IS THE SECOND PART OF FLOOR_SCRIPT. NODE KINDS ARE THOSE OF GRAPH.JSON (PI, PO, CONST, LUT, FF, HARD, GATE).

Floor	Reads	Mapping script	Netlist contents
f1-k6	read_arch k6_frac_N10_ frac_chain_mem32K_40nm	hardmap; lower; abc --lut 6	LUT6, flip-flops, and hard adder, multiplier and RAM cells that keep their Odin III type
f1-earch	read_arch of the EArch file	same as f1-k6	same kinds, EArch hard blocks
f2	read_arch (cell types only)	lower; abc --lut 6	LUT6 and flip-flops only; arithmetic is soft, so no hard node exists
f3	read_techlib lib/cmos6.o3lib	lower; abc --gates cmos6; dfflibmap cmos6	library gates (BUF, INV, NAND2, NOR2) and DFF flip-flops

where $\pi_{d,k}$ is the prior log-odds of that detector and kind; $\lambda_{k,f}(b_f)$ is the LLR of feature f in bin b_f; $w_{\phi(f)}$ weights the feature's family (structural, WL colour, NPN class or L2; the f3 calibration uses the first three only); n_{conf} and n_{ref} are the numbers of confirmed (or committed) and refuted child hypotheses (a bus slice, a register's flip-flops), which raise and lower their parent; and e_h collects fixed records (a sampled check adds $\ln 100 = 4.61$). The clip at $\pm \ln 1000 = 6.91$ limits the double counting that a naive-Bayes sum of correlated features would otherwise allow. A proof sets $L(h) \geq \ln 9999 = 9.21$ (the odds become 9999 to 1), and a counterexample sets $L(h) = -\infty$.

All three tables are *calibrated* from the training split of L0 (loki calibrate). The flow runs with default tables on every training design; each hypothesis is labelled true when it matches an L0 construct (Section III-E). The prior is the Laplace-smoothed precision of the detector, $\pi_{d,k} = \operatorname{logit} \frac{tp+1}{n+2}$ over n proposals of which tp are true. For example, the f3 comparator detector made 31,036 proposals of which 2 were true, so $\pi = \ln \frac{3}{31035} = -9.24$; the committed value is -9.244259 (experiments/f3-recovery/calib/f3.json). A bin's LLR is $\lambda = \ln \frac{(pos+1)/(POS+B)}{(neg+1)/(NEG+B)}$, with pos and neg the true and false proposals in the bin, POS and NEG the totals of the kind, and B the number of bins. The family weights w are the grid values in $\{0, 0.1, \ldots, 1\}$ that maximise the Bernoulli log-likelihood of the labels, fitted by three sweeps of coordinate ascent on out-of-fold rows (two folds, split by design). Without calibration (--calib none) the priors are 0 and the LLRs are the defaults of bayes.py. The reliability check is the expected calibration error over ten equal-width bins of the predicted probability, $\text{ECE} = \sum_b \frac{n_b}{N} |\bar{p}_b - r_b|$ with r_b the observed match rate.

Proof engines. A hypothesis with a functional claim ("these 9 LUTs compute $A + B$ at width 5") can be proved. Three engines prove claims: cone_check first when the claim has a template, the truth table otherwise or when the cone cannot be bound to the template, and transfer after an exact proof. (i) cone_check, an Odin III pass, proves that the claim's cone of level-0 cells is equivalent to a template word cell (for example $add, $sub, $eq, $mux) with its ports bound to the cone's boundary nets, under a timeout; the reference functions of the claims (claims.py) cover addition, subtraction, the six comparisons, constant comparison, 2:1 and N:1 multiplexers, counters, shift registers and LFSRs. (ii) The *truth-table* prover enumerates the claim's inputs and side nets when together they number at most 16, an exact proof or a counterexample, and otherwise simulates 4096 seeded random vectors, which can refute a claim but only raises it by $\ln 100$. (iii) *Proof transfer* groups open hypotheses by operation, parameters and the 1-WL colour of their output drivers; when one member is proved exactly, the others are tried by a lockstep cone_match of the two cones, and one transferred member per group is proved for real as a spot check, revoking the transfers of the group if it fails. A sequential claim is proved as a next-state function (state in, next state out); an FSM gets no proof.

Deciding the structure. Hypotheses that claim the same cells compete. Two hypotheses *conflict* when both kinds are exclusive, their cell sets intersect, and neither is a strict subset of the other (a strict subset is nesting: the smaller becomes a child of the larger). For each connected component C of the conflict graph the competitive probability is

$$p_{\text{comp}}(h) = \frac{e^{L(h)}}{1 + \sum_{j \in C} e^{L(j)}}, \tag{2}$$

where the "1" is the option that none of them is real; a hypothesis without conflicts has $p_{\text{comp}} = p$. Two confirmed hypotheses with equal cells are two true forms of the same logic and count as one member of C; the one whose kind ranks higher (a counter above an adder, for example) wins at commit and the other is kept as an *alternative*. A proof is spent on hypotheses with $p_{\text{comp}} \geq 0.2$, most probable first, at most 200 per round; a hypothesis below 0.01 is rejected, and re-enters when later evidence lifts it. A hypothesis is *selected for commit* when $p_{\text{comp}} \geq 0.9$, every child that could commit is already committed, and it does not conflict with a committed piece. Selection is greedy in the order (p_{comp} descending, kind rank, cell count descending, id), and the losers of a conflict become rejected or alternative.

Commit and the final structure. A selected hypothesis is rebuilt on a copy of the working design: a hypothesis is *raised* to a single word cell only if its kind is one of adder, subtractor, comparator or multiplexer, its claim has a cone_check template, and it is proved by cone_check (directly or by transfer); the cone is re-checked by cone_check unconditionally before the raise. A claim proved only by truth table is wrapped, not raised, and counters, shift registers, LFSRs,

Algorithm 1 The L1 round loop (`loki/re/loop.py`).

$level_0 \leftarrow$ read netlist as functions; $work \leftarrow level_0$
if floor f1 with hard cells **then** run `unmap` on $work$
for $r = 1$ **to** 8 **do**
  propose: every detector adds hypotheses over the view of $work$
  score: $L(h)$ by Eq. 1; **compete:** p_{comp} by Eq. 2
  prove claims with $p_{\text{comp}} \geq 0.2$: `cone_check`, truth table, transfer
  re-score and **compete**; reject $p_{\text{comp}} < 0.01$
  select $p_{\text{comp}} \geq 0.9$ (children first); **commit** by raise and submod
  if nothing committed, nothing new, $|\Delta L| \leq 10^{-6}$ **then**
    break
  end if
end for
write `design.v`, `overlay.json`, `report`; `equiv-check` against the input

FSMs, registers, constant comparisons, multiplexer trees and carry-in adders are never raised. Every selected piece except buses and roles (which only set attributes) is wrapped by `submod` in a module named for its kind, whose instance is tagged with the hypothesis id, kind, confidence p_{comp}, proof status and the recovered function. If `raise` or `submod` fails, the offending hypothesis is rejected with a record of the error and the commit is retried once without it (`reject_failed`). The structure that L1 reports is therefore exactly the set of committed pieces in `design.v`; the alternatives, rejected and open hypotheses stay in `overlay.json` with their evidence. Algorithm 1 summarises the loop. It stops when a round commits nothing, proposes nothing new and changes no log-odds by more than 10^{-6} (a fixed point), after eight rounds, or at the time budget, then flattens and lowers the final design and checks it against the input with `equiv-check`; a result of `differs` is a bug and the run exits with an error.

The `hard` detector reads the `unmap` pass's facts: a hard adder, multiplier or memory that the flow already reverse engineered is recovered directly, with the decline reasons the pass recorded (carry-in chains and the like) kept as open hypotheses. The `hard` detector does not run on `f2` or `f3`.

D. L2: graph analysis

L2 reads the L0 `graph.json` files and nothing else: never `key.json` and never the BLIF. It has three parts.

Structural features. Each node gets numeric features (fan-in, fan-out, logic level, LUT width, hard-port count), its kind and, on hard cells, its `cell_type`, plus three families: colours of 1-WL colour refinement (a hash of the node's label and the multiset of its neighbours' colours, per round, so that equal colours mean equal unfolded neighbourhoods), the NPN class of a LUT's function (invariant under permutation and negation of the inputs and the output), and the ORCA orbit counts of the 4-node graphlets around the node. Before graphlets are counted, the global signals (clock, reset, constants, global enables) are classified by role and their edges removed, so that a clock does not connect everything.

Clustering. Four methods partition a netlist graph into candidate blocks: k-means on the feature space ($k = \text{clip}(\text{round}\sqrt{N/2}, 2, 200)$), affinity propagation on a sample of at most 3000 nodes with the remaining nodes assigned to the nearest exemplar, spectral clustering (a spectral embedding of the largest connected component, then k-means), and Louvain community detection (resolution 1, no k). The first two use node features; the last two use the graph structure. A partition is scored by its *modularity* on the undirected simple projection of the graph, by its *NMI* against the known `module_path` and `construct_type` labels and by its *ARI* against `module_path` only over the LUT, flip-flop, hard and gate nodes. These two numbers are a quality report, never an input to a method: modularity says whether the clusters are dense subgraphs, NMI whether they are the designer's modules.

GNN against 1-WL. A node classifier predicts each node's `construct_type`: a two-layer GraphSAGE (GCN as the variant) of hidden width 64 with ReLU and dropout 0.2 over the graph and its reverse edges, trained with Adam (learning rate 0.01, weight decay $5 \cdot 10^{-4}$) for a fixed 100 epochs, full-batch, with class weights $1/\sqrt{\text{count}}$. The split is L0's, by design, with the class vocabulary taken from the training designs only (rarer values, and values first seen in a test design, are one class `__other__`); there is no validation set and no early stopping, so nothing is tuned on a test design. Loss and metrics cover only the LUT, flip-flop, hard and gate nodes with a known target. The question is whether message passing earns anything over what the colour refinement already knows, because a message-passing GNN is at most as discriminating as 1-WL [4]. So the 1-WL baseline predicts, for each node, the most common training class of its WL colour (backing off from round 4 to round 0 when a colour has fewer than five training nodes), and the GNN has beaten it only if its test macro-F1 is higher. Two further baselines are reported on the same nodes, the majority class and a logistic regression on the features alone. The task is that of GNN-RE [5], which maps each gate of a flat netlist to the module it belongs to; Loki's model is a baseline for the same task, not a re-implementation, and the numbers are not compared with that paper's.

E. Evaluation protocol: how L1 and L2 are scored

Gate-Breaker evaluates a stochastic model: for each design the model is sampled K times, each sample's RTL is extracted and simulated against the VerilogEval testbench, and the results are aggregated per complexity quartile as the success rate (the fraction of the K samples that pass), Success@k (the fraction of designs with a passing sample among the first k), and the AST and CFG similarity of the recovered RTL to the reference. Loki's L1 and L2 are deterministic and their output is a structure, not an RTL text, so the same four questions are asked in different terms. Table II sets the two side by side.

L1, step by step.

1) *Split.* L0 assigns each design to train or test by a hash of the design name ($\text{sha256}(\text{seed:design}) \bmod 100 < 80$

TABLE II
THE GATE-BREAKER EVALUATION AND THE LOKI EVALUATION, STEP BY STEP.

Step	Gate-Breaker	Loki
Input	cmos6 gate netlist of a VerilogEval reference, internal nets renamed `n_`⟨SHA-1 prefix⟩, ports kept	a floor netlist (`f1`, `f2`, `f3`) with names `n`k; for the neutrality check a renamed twin
Producer	a language model, K samples per design	the L1 flow, one deterministic run per design (fixed seed); no sampling
Correctness	simulate the extracted RTL against the testbench: success rate, Success@k	`equiv-check` of the flattened final design against the *input netlist*: `final_equiv` (formal; on `f1` hard cells are cut points and only soft logic is checked). A self-consistency check of the rewrite, not correctness against the reference RTL; Loki has no behavioural pass rate against the reference
Match to the truth	AST and CFG similarity to the reference RTL	each counted hypothesis against the L0 answer key: Jaccard ≥ 0.5, one-to-one; precision, recall, F1, average precision per kind and floor; hierarchy similarity; AST and CFG of `design.v` against the reference on `f3`, unconditionally (Gate-Breaker's are computed on functionally correct samples only)
Aggregate	per complexity quartile	micro-averaged per kind and floor over the test split; AST and CFG per quartile of `f3` node count

is train), so the split is fixed by the name and independent of the run.

2) *Calibrate* on the training designs of one floor (`loki calibrate`): the priors, bin LLRs, family weights and reliability curve of Section III-C.
3) *Run* `loki re` on every test netlist, calibrated and, as a diagnostic, with `--calib none`. The run never reads the answer key.
4) *Ground truth.* From `words.json`, the L0 word cells give ground-truth instances (type, F_g) with F_g the set of final cells of the word; counters, mux trees and FSMs are derived from the word connectivity.
5) *Count and match.* A hypothesis *counts* when it is committed or its $p_{\text{comp}} \geq 0.5$ (`--score-p`); it *matches* a ground-truth instance of an accepted type when the Jaccard of their cell sets is at least 0.5, greedily one-to-one by Jaccard. Precision is matched over counted, recall matched over ground truth, and average precision sweeps the threshold from 0.1 to 0.9.
6) *Buses, roles and hierarchy.* A bus matches a ground-truth bus (nodes sharing a module path and bus name, at least two bits) at the same Jaccard; its bit order is scored exactly and up to reversal. Roles are scored as precision and recall of the predicted control and datapath cells. The hierarchy similarity compares the clusters of the module paths to the committed pieces with the Dice score $F(g,h) = 2|g \cap h|/(|g| + |h|)$, $\text{HS}_{gt} = \sum_g |g| \max_h F(g,h) / \sum_g |g|$ and its converse HS_{l1}, and reports their harmonic mean.
7) *Aggregate.* `loki score` writes one `score.json` per run and `loki score-all` the floor-by-kind tables, the reliability curve of p_{comp} with its ECE, and, with `--compare`, the ablation against `--features none`.
8) *Naming neutrality.* The renamed twin of the dataset is run and scored against the original-named key; the base and renamed `score.json` must be byte-identical.

Of the Gate-Breaker metrics, the AST and CFG similarities are used on `f3` as in the paper: AST similarity is the mean, over the subtrees of the reference, of the best Jaccard similarity to a subtree of the candidate (depth 3, node kinds normalised); Gate-Breaker computes both on functionally correct samples only, Loki's are unconditional; CFG similarity is the node kernel over shortest-path histograms of the control-flow graphs. They are computed twice per design, for the gate netlist (L0-stop) and for `design.v` (L1-stop), and the L1 delta is their difference. The success rate and Success@k need samples; a deterministic flow has one attempt per design, so they are reported only for the language-model replication (Section V-E). Formal `final_equiv` does not play the role of the testbench: it compares the flattened final design with the *input netlist*, so it shows that the rewrite preserved the netlist (it is trivial on `f3`, where L1 commits nothing), not that the design is correct against the reference RTL. Loki has no behavioural pass rate against the reference.

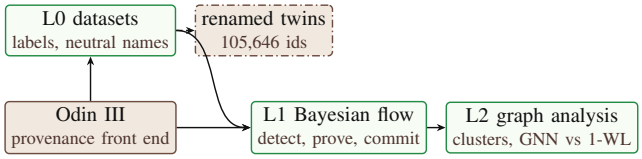


Fig. 1. The Loki III pipeline. L0 compiles the benchmarks through Odin III into provenance-tracked, labelled, anonymised datasets (and their renamed twins). L1 proposes and proves structures in a Bayesian flow that commits into the same IR. L2 reads the graphs back and scores clustering and a GNN against a 1-WL baseline. No stage has a language model.

L2, step by step. (1) The same split applies. (2) `loki analyze cluster` writes a partition per method and design, with its modularity and NMI/ARI against `module_path` and `construct_type`; the table reports the mean over designs. (3) `loki analyze gnn` fits the vocabulary and the model on the training designs and writes `metrics.json`: accuracy and macro-F1 on the training and test nodes for the GNN, the majority class, the feature-only logistic regression and the 1-WL baseline, and `beats_wl`. (4) The neutrality check re-runs both on a permuted copy of every identifier.

F. One design through the pipeline

The steps below follow one design, the 4-bit comparator micro `bm_DL_4_bit_comparator` (three comparison outputs), from source to score. Each step names the file or code that defines it.

1) *HDL to Odin III with provenance.* `read_verilog` and the common front elaborate the design. Every IR object keeps its source spans (`Node.sources()`, the only provenance the binding exposes). Before mapping, Loki snapshots every word-level cell (type, parameters, spans) and every instance site.
2) *Map to the floor.* `lower` keeps each word cell's exact span on every gate it creates; `abc -lut 6` gives each LUT the union of the spans of its cone. A LUT that merges the three comparators therefore lists all three spans.
3) *Join and label.* A final cell is matched to a word cell when the word's span set is a subset of the cell's. The primary match gives `construct_type` (here `$eq`, `$gt`, ...); a cell with several matches is flagged (`n_construct_cells` above one) and a cell with none stays `unknown`: a label is an answer key, never a guess. The module path comes from the instance spans, bus name and bit index from net names and aliases (`AeqB[0]`), and the datapath or control role from a rule table.
4) *Anonymise and write the L0 files.* Nets, cells and port bits get names `n`k from one counter. Loki writes `netlist.blif` and `netlist.v`, `graph.json` (nodes with features and labels, edges), `labels.csv` (one row per node), `words.json` (the word-level connectivity the scorer needs) and `key.json`. Only `key.json` holds original names; no analysis reads it.
5) *Verify.* Odin III's `equiv-check` compares the written BLIF with the floor's own netlist and the anonymised netlist with the original; both must read `equivalent` before the floor counts as built.
6) *L1: normalise.* `loki re` reads `netlist.blif` (never `key.json`) into a fresh Odin III design, level 0, in which every logic cell is a function (a LUT by its truth table, a gate by its library function). On `f1` with hard cells the `unmap` pass runs on the working copy and its facts feed the `hard` detector.
7) *L1: hypothesis rounds.* Each round runs the detectors, scores every hypothesis (Section III-C), proves the ones worth a proof, resolves conflicts and commits the confident ones (Algorithm 1); the next round sees the committed level.
8) *L1: commit and output.* A committed hypothesis is rebuilt as a word cell (`raise`) wrapped in a module (`submod`) tagged with its kind, confidence and recovered function. At the fixed point Loki writes `design.v`, `overlay.json` (every hypothesis with its evidence) and `report.json`, and checks the final design against the input with `equiv-check`.
9) *L2.* `loki analyze` reads `graph.json` only: structural features, four clusterings and a GNN (Section III-D).
10) *Score.* `loki score` compares the overlay with the answer key (Section III-E); `loki score-all` aggregates every run into per-kind tables.

In this micro the `f3` comparator, as re-synthesised by the minimal `.v` front, has 26 logic cells (L1 spec, Section 3); the `f2` LUTs carry the comparator spans that the join reads back.

IV. The VerilogEval benchmark and its structure

A. VerilogEval spec-to-rtl as the Gate-Breaker benchmark

Gate-Breaker evaluates on VerilogEval [2], taking the spec-to-rtl variant: 156 problems, each a natural-language prompt, a reference solution (`*_ref.sv`, module `RefModule`) and a testbench (`*_test.sv`). The Gate-Breaker flow synthesizes the reference to a cmos6 gate netlist, hands the netlist to the model, and checks the model's RTL against the testbench. Loki uses the same 156 references as L0 sources and reads the reference module as the design; the prompts and testbenches are not inputs to L0, L1 or L2. The references are SystemVerilog, read through Odin III's slang adapter (Section V-A).

B. The complexity score and its quartiles

Gate-Breaker groups the problems by a complexity score (its Eq. 2), a weighted sum of counts of HDL constructs:

$$C = 1000\,c_{\mathrm{Seq}} + 200\,(c_{\mathrm{FG}} + c_{\mathrm{F}}) + 20\,c_{\mathrm{Mo}} + 15\,(c_{@A} + c_{\mathrm{B}}) + 10\,(c_{\mathrm{M}} + c_{\mathrm{Ca}}) + 5\,c_{\mathrm{E}} + 3\,c_{\mathrm{A}} + 2\,c_{\mathrm{C}}. \quad (3)$$

The paper names the terms but does not define them, so the implementation we reuse (`complexity.py` of the second Gate-Breaker replication) fixes the reading: c_{Seq} counts clocked `always` blocks (a `posedge` or `negedge` in the sensitivity list), c_{FG} for-generate constructs, c_{F} `for` loops, c_{Mo} module instantiations, $c_{@A}$ all `always` blocks, c_{B} declarations wider than one bit, c_{M} multiplexers (ternary operators plus non-default `case` items), c_{Ca} `case` statements, c_{E} arithmetic operators (`+ - * / %`), c_{A} continuous assignments and c_{C} concatenations. A clocked block weighs 1000 on its own, so a design in which the counter recognises one scores at least 1000, which is Q3 or Q4.

The quartiles are the paper's absolute bins (its Table II), not rank quartiles of the benchmark: Q1 is 3 to 23, Q2 24 to 111, Q3 112 to 1064 and Q4 1065 to 1326. The 156 references fall 38, 39, 40 and 39 into Q1 to Q4, close to the paper's 39 per quartile. The 150 designs that build on every floor are split 38, 39, 37 and 36: the six missing ones are `Prob031`, `Prob046` and `Prob145` (Q3) and `Prob078`, `Prob151` and `Prob156` (Q4). Table III gives the counts.

Deviation: pyverilog AST against a regex fallback. The score is computed on the pyverilog AST of the first module when the file parses, and by regular-expression counts of the same terms otherwise, with the method recorded per design. Because the references are SystemVerilog, pyverilog parses only 60 of the 156; the other 96 use the regex counts. The

split is skewed by quartile (Table III): 36 of Q1 and 23 of Q2 are AST-scored, but 1 of Q3 and none of Q4. The regex counts operator characters and keywords textually, so they differ from an AST count (the observed Q4 range, 1069 to 3259, exceeds the paper's bin top of 1326 for this reason or because the problems differ; we do not separate the two). The regex also never counts `always_comb`, `always_ff` or `always_latch` blocks, which 34 references use, so `c@A` and `cSeq` are undercounted in the 96 regex-scored designs, most of Q3 and Q4; the quartile split and the feature trends below rest on approximate scores. The 156 scores equal those of the replication that computed them first, exactly, so the quartiles here and in Table XII are the same partition.

TABLE III
THE VERILOGEVAL SPEC-TO-RTL REFERENCES BY GATE-BREAKER COMPLEXITY QUARTILE (ABSOLUTE BINS OF TABLE II; SCORE RANGE OBSERVED IN THE 156 REFS; PYVERILOG AST SCORES AGAINST REGEX-FALLBACK SCORES; BUILT = DESIGNS WITH ALL THREE FLOORS).

Quartile	Bin	Observed	Refs	AST/regex	Built
Q1	3–23	3–23	38	36/2	38
Q2	24–111	24–90	39	23/16	39
Q3	112–1064	120–1060	40	1/39	37
Q4	1065–1326	1069–3259	39	0/39	36
All			156	60/96	150

C. The feature matrix

Gate-Breaker characterises its benchmark by the HDL structures it contains. We build the same kind of matrix for the 150 built designs, from the `f3` floor of each, with rules fixed in advance (`summarize.py`): *arithmetic* counts the word cells of type `$add $sub $mul $div $mod $neg $pos $pow $alu $macc`; *select/mux* those of type `$mux $pmux $bmux $demux`; *storage* the register words (`$dff`, `$dffe`, `$adff`, `$adffe`, `$sdff`, `$sdffe`, `$sdffce`, `$dffsr`, `$aldff`, `$dlatch`, `$adlatch`); *FSM* is 1 when Odin III's `fsm` pass marked any word; *control* is 1 when any label row has role `control` or `mixed`; and *wide buses* counts the distinct bus names with a bit index of at least 1. The matrix uses one symbol per cell: ✗ for count 0, ✓ for one, ✓✓ for two or more; FSM and control are 0/1 flags, so they show ✗ or ✓ only.

The bus column is `buses_wide`, not `buses`. The latter counts every named signal, including single-bit ports, so it is at least 1 for all 150 designs and at least 2 for 147, and would put a check mark in every row. The wide count is the quantity a "multi-bit bus" row means; both columns are committed in `features.csv`.

Table IV shows 12 designs, three per quartile at fixed ranks by size, so the sample is not hand-picked; Table V counts, per quartile, the designs that show each feature. The storage trend is partly tautological: a clocked block adds 1000 to the score (Eq. 3) and storage counts register words, so storage rises with the quartile by construction; the other trends are conditioned on the score as well. Storage is absent from Q1 and Q2 (0 of 77 designs) and present in 31 of 37 Q3 and all 36 Q4 designs. The other features grow with the quartile: select/mux 8% in Q1 to 56% in Q4, control 3% to 83%, FSMs 0 of 77 below Q3 and 15 of 36 in Q4. Arithmetic is rare throughout (21 of 150, 14%), so the benchmark leans on control, selection and storage more than on arithmetic datapaths. Wide buses are common from Q2 up (90%, 59%, 100%) but nearly absent in Q1 (8%), whose members mostly have only single-bit signals. The FSM flag is Odin III's detection, not a reading of the source: `Prob107_fsm1s` in Table IV has storage and control cells but no FSM mark.

TABLE IV
BENCHMARK FEATURE MATRIX, A SAMPLE OF 12 OF THE 150 DESIGNS (THE DESIGNS AT RANKS $\lfloor n/6 \rfloor$, $\lfloor n/2 \rfloor$ AND $\lfloor 5n/6 \rfloor$ BY F3 NODE COUNT INSIDE EACH QUARTILE). ✗: FEATURE ABSENT; ✓: ONE; ✓✓: MORE THAN ONE; FSM AND CONTROL ARE 0/1 FLAGS. FEATURES ARE READ FROM THE F3 FLOOR (SECTION IV-C).

Design	Q	f3 nodes	Arith	Sel	Stor	FSM	Ctl	Bus
Prob028_m2014_q4a	Q1	3	✗	✗	✗	✗	✗	✗
Prob036_ringer	Q1	8	✗	✗	✗	✗	✗	✗
Prob091_2012_q2b	Q1	17	✗	✗	✗	✗	✗	✓
Prob125_kmap3	Q2	15	✗	✗	✗	✗	✗	✗
Prob015_vector1	Q2	32	✗	✗	✗	✗	✗	✓✓
Prob123_bugs_addsubz	Q2	166	✓✓	✓	✗	✗	✗	✓✓
Prob107_fsm1s	Q3	11	✗	✓	✓	✗	✓	✗
Prob117_circuit9	Q3	28	✓	✓	✓✓	✗	✓	✓
Prob045_edgedetect2	Q3	89	✗	✗	✓✓	✗	✗	✓✓
Prob128_fsm_ps2	Q4	30	✗	✗	✓	✓	✓	✓✓
Prob140_fsm_hdlc	Q4	52	✗	✗	✓	✓	✓	✓
Prob115_shift18	Q4	1121	✗	✓✓	✓✓	✗	✓	✓✓

D. Benchmark size per technology mapping

Table VI gives the size of the 150 designs on each floor. The averages hide a heavy tail. Over all designs the mean node count is 163.5 on `f1-k6`, 111.4 on `f2` and 362.0 on `f3`, but the medians are 14.0, 13.5 and 25.0: most references are small (125 of 150 have at most 100 nodes at `f3`) and a few are large (9 exceed 1000; the largest, `Prob144_conwaylife`, has 15,487). The gate mapping multiplies the node count of the LUT mapping by about 3 on the mean and 1.9 at the median (362.0 against 111.4 on average, 25.0 against 13.5 at the median), as expected when a LUT of up to six inputs becomes a network of cells from a library of one- and two-input gates (BUF, INV, NAND2, NOR2). `f1-k6` sits above `f2` on average and in Q3 and Q4 (109.4 against 77.4; 453.9 against 269.8) while the two coincide in Q1 and Q2 (6.3 and 6.1; 100.2 and 100.0): the `f1-k6` floor keeps hard cells (21 designs hold hard adders) where `f2` spreads arithmetic over LUTs, which is consistent with the excess but is not isolated here. By quartile on `f3`, the mean grows from 9.9 nodes in Q1 to 972.1 in Q4, but the Q4 median is only 51.5: the complexity score tracks the number and kind of HDL constructs, not the netlist size. For scale, the micro suite's `f3` floors hold 50,047 nodes over 65 designs, a mean of about 770.

TABLE V

DESIGNS SHOWING EACH FEATURE, PER COMPLEXITY QUARTILE: COUNT (PERCENT OF THE QUARTILE). THE LAST COLUMN COUNTS DESIGNS WITH TWO OR MORE INSTANCES OF THE FEATURE (ALWAYS 0 FOR THE 0/1 FLAGS). WIDE BUSES ARE DISTINCT MULTI-BIT BUSES; A DESIGN WITH ONLY SINGLE-BIT PORTS HAS NONE.

Feature	Q1 (n=38)	Q2 (n=39)	Q3 (n=37)	Q4 (n=36)	All (n=150)	≥ 2
Arithmetic	2 (5%)	4 (10%)	7 (19%)	8 (22%)	21 (14%)	9
Select/mux	3 (8%)	10 (26%)	12 (32%)	20 (56%)	45 (30%)	12
Storage	0 (0%)	0 (0%)	31 (84%)	36 (100%)	67 (45%)	12
FSM	0 (0%)	0 (0%)	1 (3%)	15 (42%)	16 (11%)	0
Control	1 (3%)	4 (10%)	15 (41%)	30 (83%)	50 (33%)	0
Wide buses	3 (8%)	35 (90%)	22 (59%)	36 (100%)	96 (64%)	55

TABLE VI

BENCHMARK SIZE PER TECHNOLOGY-MAPPING FLOOR, 150 DESIGNS: MEAN / MEDIAN NODE COUNT (TOP), AND PER COMPLEXITY QUARTILE (BOTTOM; NODES). NODES ARE ALL GRAPH NODES OF THE FLOOR NETLIST: PRIMARY INPUTS AND OUTPUTS, CONSTANTS, CELLS.

All 150 designs	f1-k6	f2	f3
Nodes	163.5 / 14.0	111.4 / 13.5	362.0 / 25.0
Cells	114.6 / 5.0	62.5 / 4.0	313.2 / 13.5
Nets	218.3 / 12.0	93.5 / 10.0	344.2 / 21.0
Nodes by quartile	**f1-k6**	**f2**	**f3**
Q1 (n=38)	6.3 / 5.0	6.1 / 5.0	9.9 / 8.0
Q2 (n=39)	100.2 / 26.0	100.0 / 26.0	238.2 / 32.0
Q3 (n=37)	109.4 / 18.0	77.4 / 13.0	260.6 / 28.0
Q4 (n=36)	453.9 / 26.5	269.8 / 26.5	972.1 / 51.5

TABLE VII

THE L0 CEILING ON `f3`, MICRO SUITE AND VERILOGEVAL REFERENCES. BOTH COLUMNS ARE COUNTS OVER ALL BUILT DESIGNS; THE LAST ROW IS THE PER-DESIGN COVERAGE OF DECLARED SIGNALS.

	Micro	VerilogEval
Source designs	68	156
Built on `f3`	65	150
Refused by the reader	3	2
Failed equivalence gate	0	4
Gate nodes, each with a label row	50,047	54,306
Rows with `cell_type` and HDL span	100%	100%
Module scopes / reference modules	112 / 113	150 / 150
Leaf signal names / declared signals	611 / 666	841 / 745
Per-design signal coverage: median	100%	100%
minimum	55.6%	75.0%
geometric mean	–	0.986

V. RESULTS

A. L0 on VerilogEval: the provenance ceiling

The provenance experiment is run on the 156 VerilogEval spec-to-rtl reference designs (Section IV) next to the micro suite. The question is the one L0 exists to answer: how much of the reference does the answer key reconstruct, per floor, with no recovery step at all. The experiment builds the three floors `f1-k6`, `f2` and `f3` for every reference (`f1-earch` is not built; one architecture is enough because `f2` and `f3` do not depend on it).

Coverage of the build. Of the 156 references, 150 build on all three floors, and `loki check` reports zero errors on the assembled dataset. Two are skipped by Odin III's SystemVerilog reader: `Prob151_review2015_fsm` and `Prob156_review2015_fancytimer` (both Q4), with `ODIN3_ERR_PARSE ... an explicit 4-bit cast is not supported`. Four fail Loki's own equivalence gate and are recorded as failed rather than forced in: `Prob031_dff` (the reference initialises `q` to `1'hx`; the writer-versus-Odin III BLIF check reports `differs` on `f1` and `f2`, while `f3` was equivalent) and `Prob046_dff8p`, `Prob078_dualedge` and `Prob145_circuit8`, whose negedge flip-flops make `equiv-check` return an error status for both comparisons. One design, `Prob144_conwaylife`, was killed by Wyrd's default 2 GB step memory limit (peak 4.3 GB) and built after a rerun with an 8 GB limit. Reading `.sv` sources needed two fixes in Loki (SystemVerilog is read with `--nolatches` only, since `--vtr-compat` is refused for it, and the slang adapter is registered with the process); both are in `loki/odin.py` and apply to the experiment run.

The ceiling. On `f3` the 150 designs have 54,306 gate nodes. Every node has a label row, a `cell_type` and an HDL span: the label, `cell_type` and HDL coverage have geometric mean 1.000 over designs. Label coverage is rows over nodes, so it shows that a row exists, not that its fields are informative; unmatched cells stay `unknown`. The 150 module scopes of the 150 reference modules are all present (module coverage 1.000). Signal coverage, the recovered leaf signal names over the signals the reference declares, has geometric mean 0.986 (arithmetic mean 1.029): 120 designs are at exactly 1, 28 are below it, and 2 are above it, where flattened array elements add names. The micro suite shows the same shape with a slightly lower ceiling (Table VII): 99.1% of module scopes and 91.7% of signal names (an aggregate ratio, where VerilogEval's 841 over 745 is inflated by those two designs; capped at each design's declared count it is 715 over 745, 96.0%). The two sets agree on the message of Section V-C: every node has a label row, a `cell_type` and an HDL span, and what the mapping erased is the signal names, not the label rows. (Label coverage counts rows, not informative values: a cell that no rule matched keeps the value `unknown`.)

Caveats, reported with the numbers. (i) `--nolatches` turns inferred-latch paths into don't-care and `lower --async sync` turns asynchronous resets into synchronous clears, so a built netlist can differ from the reference's behaviour even though Loki's own checks (netlist against its

TABLE VIII

L1 RECOVERY ON THE RENAMED FLOORS, PER KIND. PRECISION/RECALL/F1 AGAINST THE L0 GROUND TRUTH; CALIBRATED (MICRO-FIT) COLUMNS. THE HARD KINDS RECOVER FROM `UNMAP` FACTS; THE REST MUST BE RE-FOUND FROM THE GATES.

Kind (f1)	Precision	Recall	F1
hard_add	0.985	1.000	0.993
hard_mul	1.000	1.000	1.000
ff_en	–	–	0.364
register	–	–	0.021
mux	–	–	0.018
mux_bit	–	–	0.081
macro-F1 (all kinds)			0.354
Kind (f2)			
adder	1.000	0.134	0.237
subtractor	1.000	0.074	–
mux_bit	–	–	0.100
adder_cin	–	–	0.029
macro-F1 (all kinds)			0.108

TABLE IX

F3 L1 PER KIND ON THE 15 TEST DESIGNS. THE CALIBRATED COLUMNS ARE THE PROTOCOL'S SETTING; THE `--CALIB NONE` COLUMNS ARE A DIAGNOSTIC OF WHAT THE DETECTORS PROPOSE BEFORE CALIBRATION PRUNES IT. A KIND WITH NO COUNTED PREDICTION HAS NO F1 AND IS LEFT OUT OF THE MACRO AVERAGE.

		calibrated			`--calib none`			
Kind	GT	counted	matched	F1	counted	matched	precision	F1
adder	17	0	0	–	1	1	1.000	0.111
adder_cin	17	0	0	–	0	0	–	–
comparator	20	0	0	–	14	0	0.000	0.000
eq_const	16	0	0	–	1,141	0	0.000	0.000
ff_srst	2	0	0	–	7	0	0.000	0.000
hard_add	17	0	0	–	0	0	–	–
hard_mul	11	0	0	–	0	0	–	–
mux	7	0	0	–	0	0	–	–
mux_bit	7	0	0	–	0	0	–	–
register	22	0	0	–	3	1	0.333	0.080
subtractor	8	0	0	–	0	0	–	–
macro-F1	144	0	0	0.000	1,166	2		0.038
buses	120	2,569	0	0.000	2,835	0	0.000	0.000
datapath role	20,541	20,541	20,541	1.000	20,541	20,541	1.000	1.000
control role	0	0	0	–	0	0	–	–

anonymised twin and against Odin III's writer) pass; the build gate guarantees internal consistency, not equality with the reference. (ii) The reference module, instance and signal counts come from pyverilog where it parses the reference (130 of the 150) and from a regex fallback otherwise (20); the regex counts are approximate and the reference signal count is the weakest number in this section. (iii) The same pyverilog limit decides the complexity quartiles (Section IV-B). (iv) A train/test split by design is fixed by the manifest rule (117 train, 33 test on `f3`); it is not used here, where no model is fitted.

B. L1 recovery: what the flow already reverse engineered, and what it erased

Table VIII reports L1 on the renamed floors (identical to base, as the neutrality check requires). The head-line result is the `hard` detector: hard adders recover 67 of 67 with precision 0.985, hard multipliers 30 of 30, because the synthesis flow itself already reverse engineered them and `unmap` says so. The rest is the honest difficulty: on f1 the calibrated macro-F1 is 0.354 (counted 271, matched 107, ground truth 688), with `ff_en` at 0.364, registers at 0.021 and muxes at 0.018; on f2 it is 0.108 (91/19/686), with adders at F1 0.237 (precision 1.000, recall 0.134) and most other kinds near zero. Without calibration the flow proposes far more and confirms little: f1 macro-F1 0.202 with 1,128 counted (323 of them declined-adder proposals), f2 0.075. Calibration is honest about itself: ECE 0.217 calibrated against 0.650 uncalibrated.

C. f3: recovery from the cmos6 gate netlist

Gate-Breaker's own input is a cmos6 gate netlist, so we built the f3 floor (65 of 68 micro designs; the gate reader refuses three hard-block sources) and asked the same questions of it. The L0 answer key is complete at this level in that all 50,047 gate nodes carry a label row, a cmos6 `cell_type` and an HDL span. What provenance reconstructs of the reference is nearly complete but not total: 112 module scopes against 113 module definitions (99.1%; 54 of 65 designs complete) and 611 distinct leaf signal names against 666 declared signals (91.7%; median design 100%, minimum 55.6%). The module gap is reference-side (generate branches and unused submodules never elaborate); the signal gap is synthesis-side (nets that mapping created or merged have no original name left). The ceiling is consequently what the mapping preserved, not the source RTL.

Table IX reports L1 on the 15 f3 test designs (23,098 gate nodes), base and renamed, with a calibration fitted on the 50 train designs' f3 floors. The result is the strict one: no construct is counted (0 of 144 ground-truth instances; macro-F1 0.0) and no hypothesis is committed at all. The f3 priors are strongly negative because the gate-level detectors are imprecise on the train floors (comparator -9.24 from 31,036 proposals with 2 true), so the best calibrated confidence in any test design is a bus at $p = 0.764$, below the 0.9 commit threshold, and the L1-stop Verilog is byte-identical to the input netlist in 15 of 15 designs. Without calibration the detectors count 1,166 constructs for 2 matches (macro-F1 0.038; one adder, one register) and 2,835 buses for 0 matches. Buses are proposed but never reconstructed: 2,569 counted bus hypotheses against 120 ground-truth buses, 0 matched, so no width or bit order is recovered. The role detector labels every logic cell datapath, which is 100% correct only because the f3 answer key holds no control cells (0 of 50,047; f1 has 4): control-versus-data is not answerable from f3 provenance, and the datapath F1 of 1.000 over 20,541 cells is vacuous. The hard detector stays f1-only by design (`unmap` runs only on f1 floors with hard cells), so the 28 hard ground-truth instances are unreachable here. Base and renamed `score.json` digests are identical (9368cfc26303): naming contributes no decision bit at f3 either.

The AST and CFG deltas (Table X) follow from the zero-commit result: with the f3 calibration L1-stop is byte-identical to L0-stop, so both metrics have delta exactly 0.0000 in every quartile (AST over 15 designs; CFG over the 13 whose CFG

TABLE X

F3 AST (GATE-BREAKER ALGORITHM 2, DEPTH 3, PYVERILOG) AND CFG (ALGORITHM 3, ODIN3 PROBE) SIMILARITY AGAINST THE MICRO REFERENCE RTL, L0-STOP (GATE NETLIST) AGAINST L1-STOP (`DESIGN.V`); QUARTILES BY F3 NODE COUNT. WITH THE F3 CALIBRATION L1-STOP EQUALS L0-STOP, SO EVERY DELTA IS 0.0000; THE TABLE THEREFORE SHOWS THE L0 VALUES AND THE DIAGNOSTIC (`--CALIB NONE`) DELTAS.

Quartile	AST L0	AST L1	ΔAST	CFG L0	CFG L1	ΔCFG
0 (4 designs)	0.4428	0.4428	0.0000	0.7909	0.7909	0.0000
1 (4)	0.3765	0.3765	0.0000	0.4224	0.4224	0.0000
2 (4)	0.3969	0.3969	0.0000	0.7013	0.7013	0.0000
3 (3 AST / 1 CFG)	0.4003	0.4003	0.0000	0.7507	0.7507	0.0000
mean (15 / 13)	0.4044	0.4044	0.0000	0.6468	0.6468	0.0000
diagnostic `--calib none` (L1 commits hierarchy)						
0 (4)	0.4428	0.4277	−0.0151	0.7909	0.7776	−0.0133
1 (4)	0.3765	0.3756	−0.0009	0.4224	0.4099	−0.0124
2 (4)	0.3969	0.4542	+0.0573	0.7013	0.7014	+0.0001
3 (3 AST / 1 CFG)	0.4003	0.4007	+0.0004	0.7507	0.7561	+0.0055
mean	0.4044	0.4155	+0.0111	0.6468	0.6394	−0.0075

the probe can finish, 12,000 nodes being its practical $O(n^2)$ limit). The uncalibrated diagnostic, where L1 commits 1,535 hypotheses in 11 of 15 designs, moves the means by +0.0111 (AST) and -0.0075 (CFG), with mixed per-quartile signs; the commits are mostly wrong `eq_const` and bus pieces, so this is not evidence about recovery quality. A flatten control (Odin III `proc; flatten`, which also lowers the committed processes) inlines the committed hierarchy and raises both metrics over the gate netlist: AST 0.4576 against 0.4044 (+0.0533, all 15 designs) and CFG 0.6941 against 0.6382 (+0.0559, the 12 designs the probe can score). Gate-Breaker's Algorithms 2 and 3 read a single module, while Loki's L1 commits a hierarchy, so a recovery that produces hierarchy is systematically penalised until it is flattened: for a netlist-to-RTL benchmark this is a metric limitation, not a property of the recovery.

D. L2: clusters that carry the hierarchy, and a GNN that beats 1-WL

Table XI scores the four clusterings against the known module hierarchy. Louvain leads (modularity 0.4220, NMI 0.1360); k-means and affinity propagation do not find the hierarchy (negative modularity), which is itself a result: graph structure alone, without the labels, favours the community methods. 271 of 272 partition records are identical between base and renamed floors; the one difference exposed a determinism defect in the spectral path (repeated runs on the same base design give modularities -0.0079, -0.0410 and 0.0252), reported as an open issue rather than hidden.

The GNN result answers the paper's question directly. A node-only GraphSAGE loses to the 1-WL baseline on real micros (macro-F1 0.2413 against 0.2711); combining node, graphlet, WL and NPN features turns it around (0.3545 against 0.2711 on the earlier real-micro run), and the full labelled run of this experiment gives test macro-F1 0.4241 against 0.2717 (accuracy 0.8863 against 0.8492, 3,932 test nodes across 13 designs). The features carry the recovery, not the message passing alone.

TABLE XI

L2 CLUSTERING AGAINST THE KNOWN MODULE HIERARCHY (68 DESIGNS, MEAN OVER THE RENAMED FLOORS, IDENTICAL TO BASE). MODULARITY AND NMI(MODULE_PATH) PER METHOD.

Method	Modularity	NMI
Louvain	0.4220	0.1360
Spectral	0.3620	0.1378
Affinity propagation	-0.2264	0.2708
k-means	-0.2912	0.3220

TABLE XII

GATE-BREAKER METRICS ON THE VERILOGEVAL QUARTILES (ONE RUN, K=10, 4 WORKERS, 180 S PER CALL, 7,440 S WALL CLOCK, 740 CALLS; DELTAS VS THE PAPER'S O3-MINI IN THE TEXT).

Quartile (n)	SR	S@1	S@5	S@10	AST	CFG
Q1 (36)	0.903	0.944	1.000	1.000	0.894	0.942
Q2 (37)	0.778	0.811	1.000	1.000	0.761	0.890

E. The Gate-Breaker metrics

Gate-Breaker's metrics compare a recovered RTL against a reference: Success Rate, Success@k, AST similarity and CFG similarity. Loki recovers structure, not a new RTL text, so the direct comparison is the replication: the Gate-Breaker harness itself, run on the simple benchmarks with the only language model available on the host, gave success rate 0.933 on Q1 and 0.829 on Q2, Success@10 1.000 in both quartiles, AST similarity 0.576 and 0.444, CFG similarity 1.000 and 0.736, with 169 of 174 extracted candidates functionally correct – a first replication on the fallback micro set, not the paper's VerilogEval set. The VerilogEval replication, on the quartiles the paper used, is reported in Table XII with the AST and CFG kernels recomputed with common-width feature vectors (the first run padded per graph and truncated in the zip, which the v1.1.0 CFG probe found; those columns are not comparable to it).

Seventy-four of the 78 Q1+Q2 designs are usable (three excluded because their synthesized netlists fail the official testbench self-check, one because its 241 KB prompt exceeds the OS argument limit); every scored design is solved by attempt 5. The deltas against the paper's O3-Mini: Q1 success rate -0.092, AST -0.030, CFG -0.011; Q2 success rate -0.039, Success@5 +0.125, Success@10 +0.111, AST -0.164, CFG +0.083. The Q1 gap is harness no-output, not wrong RTL; the Q2 AST gap is truth-table and case references against the model's boolean expressions. Two caveats: the host's only model path routes through a gateway that aliases a DeepSeek backend as Claude Opus (so the model column is not verifiably the paper's comparison class), and the CLI exposes none of the paper's sampling flags (T=0.85, p=0.95). The CFG column uses the common-width node kernel (the odin3 CFG probe); the AST column uses the paper's pyverilog Algorithm 2 of Gate-Breaker.

What the two lines of work measure is complementary. The language model recovers functional behaviour from the gates and names nothing reliably; Loki recovers the structures the

TABLE XIII

ROUND-2 L1 (RQ-L1): RECOVERY FROM THE F3 GATE NETLIST RE-MAPPED TO LUT6 AGAINST THE NATIVE F2 FLOOR, SAME 65 MICRO DESIGNS. `TEST` IS THE 15 HELD-OUT DESIGNS, `ALL` THE 65. CALIBRATION: THE MICRO-FIT F2 TABLE (GATE ROWS: THE F3 TABLE). HARD KINDS EXCLUDED FROM COUNTS AND AVERAGES; MACRO-F1 AVERAGES THE KINDS WITH A COUNTED PREDICTION.

Netlist	Calib.	Split	Runs	Counted	Matched	GT	micro-F1	macro-F1
f3 gate netlist	calibrated	test	15	0	0	116	0.000	0.000
	uncalibrated	test	15	1,166	2	116	0.003	0.038
f3 re-mapped to LUT6	calibrated	test	15	68	2	116	0.022	0.076
	calibrated	all	65	92	14	580	0.042	0.062
	uncalibrated	test	15	252	4	116	0.022	0.044
	uncalibrated	all	65	921	31	580	0.041	0.049
f2 floor (native LUT6)	calibrated	test	15	72	4	116	0.043	0.153
	calibrated	all	65	88	16	579	0.048	0.095
	uncalibrated	test	15	255	13	116	0.070	0.121
	uncalibrated	all	65	930	32	579	0.042	0.070

TABLE XIV

OPERATOR-RECOVERY F1 (RQ-L7) OF THE EXISTING AND NEW L1 RUNS: THE MULTISET OF RECOVERED OPERATORS (CLASS AND OPERAND WIDTH) AGAINST THE OPERATORS OF THE L0 KEY, POOLED OVER DESIGNS. `TYPE` IGNORES THE WIDTH, `TW` REQUIRES IT.

Run set	Designs	Key ops	Got ops	P	R	type-F1	tw-F1
f1-k6, calibrated	68	468	169	0.657	0.237	0.349	0.204
f1-k6, none	68	468	964	0.233	0.481	0.314	0.166
f2, calibrated	68	468	21	0.810	0.036	0.070	0.061
f2, none	68	468	756	0.180	0.291	0.222	0.080
f3 gates, calibrated	15	108	0	–	0.000	0.000	0.000
f3 gates, none	15	108	1,161	0.009	0.102	0.017	0.008
f3 as LUT6, calibrated	65	463	19	0.895	0.037	0.071	0.062
f3 as LUT6, none	65	463	753	0.190	0.309	0.235	0.086
VerilogEval f2, calibrated	150	3,585	11	0.727	0.002	0.004	0.002
VerilogEval f2, none	150	3,585	1,151	0.196	0.063	0.095	0.042
VerilogEval f3, calibrated	150	3,585	2	1.000	0.001	0.001	0.001
VerilogEval f3, none	150	3,585	5,488	0.107	0.163	0.129	0.027

flow preserved (hard blocks, the hierarchy, the signal roles its features see) and proves what it claims. The Gate-Breaker future work – bus naming, datapath/control classification, partitioning – is exactly L2's table, and its numbers are the baseline a model-assisted Loki would have to beat.

F. Round 2: five follow-up questions

The first round left five questions open. Each carries a hypothesis and a decision rule; the thresholds were set by the implementer during the round and are not time-stamped before the data (Appendix A); a later round should fix them in a commit before running. All five are compute-only: no language-model call was made, and the language-model figures are the round-1 Gate-Breaker replication, read again. Every number in this subsection is a macro generated from the summary tables of `experiments/round2`.

RQ-L1: is the f3 failure a vocabulary problem? The L1 detectors read LUT covers, and the f3 netlist is a cmos6 gate netlist. H1: re-mapping the same gates into LUT6 recovers close to the f2 rate. We read each f3 `netlist.blif` into Odin III, run `abc --lut 6`, and give the result to `loki re` as an f2 netlist. Odin III drops the cell names in `abc`, so each of the 8,712 LUTs is expanded back to the gates it covers: a bit-parallel simulation signature matches its output to the f3 net of the same function, and the cover is the cone of that net down to the matched inputs. 0.230 of the LUTs ABC resynthesised implement a function that is no f3 net; they take the support-

TABLE XV

AST AND CFG SIMILARITY AGAINST OPERATOR F1 ON THE 15 F3 TEST DESIGNS (THE F3-RECOVERY RUNS). CALIBRATED AND UNCALIBRATED L1 TIE ON AST AND CFG; THE FLATTEN CONTROL MOVES BOTH AND LEAVES THE RECOVERY RECORD UNCHANGED.

Condition	AST	CFG	Ops recovered	type-F1
L0-stop gate netlist	0.4044	0.6468	0	0.0000
L1 calibrated	0.4044	0.6468	0	0.0000
L1 uncalibrated	0.4155	0.6394	1,161	0.0173
L1 uncalibrated, flattened	0.4576	0.6941	1,161	0.0173

TABLE XVI

HEAD-TO-HEAD ON THE VERILOGEVAL DESIGNS THE LANGUAGE MODEL WAS SCORED ON (Q1 AND Q2, FUNCTIONALLY CORRECT SAMPLES; AST DEPTH 3 AND CFG AS IN TABLE XII). LOKI COLUMNS: THE OUTPUT VERILOG OF THE FLOW AT THAT FLOOR AGAINST THE REFERENCE `.sv`; `L0` IS THE NETLIST BEFORE L1.

Metric	LLM (round 1)	f3 netlist (L0)	f3 L1 cal.	f3 L1 none	f2 L1 cal.	f2 L1 none
Q1 AST	0.894	0.763	0.763	0.680	0.804	0.720
Q1 CFG	0.942	0.751	0.751	0.565	0.821	0.664
Q2 AST	0.761	0.460	0.460	0.438	0.459	0.448
Q2 CFG	0.890	0.743	0.743	0.679	0.787	0.764
Q1+Q2 AST	0.827	0.611	0.611	0.559	0.632	0.584
Q1+Q2 CFG	0.916	0.747	0.747	0.622	0.804	0.714

contained share of their consumers' covers, and 153 are left with an empty cover. The expanded hypotheses are then scored against the unchanged f3 key, and the native f2 floor of the same 65 designs, with the same calibration, is the comparison (Table XIII). On the gate netlist the calibrated flow matches 0 of 116 instances in the 15 test designs. After the LUT6 re-map it matches 2 (native f2: 4), and over all 65 designs 14 against 16 of 580, a micro-F1 ratio of 0.87 in-sample (the 65 designs include the designs the f2 calibration was fitted on) against 0.51 on the held-out test split. The rule asked for a ratio of at least 0.8 on both splits, so H1 is inconclusive. The in-sample recovery is that of f2, on the same kinds (adders, subtractors, muxes), but the 15-design test split holds only a handful of matches (2 against 4), so its ratio is noise in either direction. The result is consistent with the f3 failure being the detectors' vocabulary and not missing information in the gates, but does not show it; the f2 level it would recover to is itself low.

RQ-L7: are AST and CFG blind to what Loki recovers? We score recovery with a word-level operator F1. The L0 key lists the word cells of the elaborated reference; a run's recovered operators are its counted hypotheses (committed, or $p_{\mathrm{comp}} \geq 0.5$), mapped to an operator class and an operand width (add, sub, mul, eq, ne, lt, le, gt, ge, mux, register). Precision and recall are multiset overlaps, so the score does not depend on which cells carry an operator; `type` ignores the width and `tw` requires it. Table XIV rescores every run set. The floors separate cleanly, which neither AST nor CFG can show because both exist only on f3: type-F1 0.349 on f1 (the hard detector), 0.070 on f2 and 0.000 on the f3 gates. On the f3 test designs (Table XV) the calibrated and uncalibrated L1 runs tie on both metrics, AST by 0.0111 and CFG by -0.0075, while operator type-F1 moves by 0.0173; the flatten control raises AST by 0.0422 and CFG by 0.0547 and leaves the recovery

TABLE XVII

LOKI ON ALL 150 VERILOGEVAL DESIGNS BY COMPLEXITY QUARTILE (MEAN AST AND CFG OF THE OUTPUT VERILOG AGAINST THE REFERENCE; LAST COLUMN: OPERATOR TYPE-F1 OF THE UNCALIBRATED F2 RUN). CALIBRATED F3 EQUALS THE F3 NETLIST WHEREVER L1 COMMITS NOTHING.

Quartile	n	f3 AST	f3 CFG	f3 cal. AST	f3 cal. CFG	f2 cal. AST	f2 cal. CFG	op F1
Q1	38	0.754	0.750	0.754	0.750	0.793	0.820	0.061
Q2	38	0.455	0.741	0.455	0.741	0.455	0.784	0.075
Q3	32	0.614	0.751	0.605	0.750	0.607	0.811	–
Q4	21	0.505	0.768	0.505	0.768	0.507	0.870	–

TABLE XVIII

EVIDENCE-FAMILY ABLATION (RQ-L4) ON THE 49 HELD-OUT F2 DESIGNS (16 MICRO, 33 VERILOGEVAL). EACH ROW REFITS THE CALIBRATION ON THE 169 TRAIN DESIGNS WITH THE FAMILY REMOVED (ITS WEIGHT SET TO 0). THE L2 FAMILY HAS NO FEATURE IN LOKI RE AND IS NOT APPLICABLE. THE INTERVAL IS A 95% DESIGN BOOTSTRAP OF MICRO-F1 (HARD KINDS EXCLUDED); THE NEXT COLUMN IS MICRO-F1 WITH THE DESIGN VE_PROB153_GSHARE (ABOUT 1,300 OF THE 1,558 INSTANCES) LEFT OUT.

Calibration	Counted	Matched	micro-F1	95% interval	w/o gshare	macro-F1	ECE	AUC
defaults (no calibration)	1,708	26	0.0159	0.0036–0.0912	0.0553	0.113	0.841	0.437
full calibration	10	3	0.0038	0.0000–0.0444	0.0225	0.145	0.061	0.885
without npn	10	3	0.0038	0.0000–0.0444	0.0225	0.145	0.061	0.885
without wl	13	3	0.0038	0.0000–0.0435	0.0222	0.145	0.024	0.930
without struct	11	4	0.0051	0.0000–0.0578	0.0299	0.163	0.056	0.884
priors only	14	4	0.0051	0.0000–0.0574	0.0295	0.163	0.026	0.918

TABLE XIX

GRAPH SANDWICH (RQ-L5): SPEARMAN CORRELATION OF THE BRACKET WIDTH (CANDIDATE FRACTION OF THE CELLS) AND OF THE COMMITTED FRACTION WITH THE PER-DESIGN F1; CELL-LEVEL SOUNDNESS AGAINST THE UNION OF THE SCORED GROUND-TRUTH FOOTPRINTS (MUST: COMMITTED CELLS; MAY: COMMITTED PLUS CANDIDATE CELLS).

Run set	Designs	ρ width	p	ρ must	must prec.	must rec.	may rec.
VerilogEval f2, uncalibrated	90	0.087	0.417	0.485	0.840	0.240	0.625
VerilogEval f2, calibrated	90	−0.203	0.055	0.819	0.821	0.025	0.188
VerilogEval f2, calibrated, rp 0.1	90	−0.346	0.001	0.819	0.821	0.025	0.447
VerilogEval f3, uncalibrated	90	−0.124	0.243	0.168	0.792	0.347	0.727
micro f2, uncalibrated	48	0.162	0.272	0.219	0.879	0.268	0.364
micro f2, calibrated	48	0.052	0.725	0.844	0.932	0.021	0.132

TABLE XX

STRUCTURAL PREFILTER (RQ-L5B), MEASURED OFFLINE ON THE 210 CONE_CHECK CALLS OF ALL ROUND-2 RUNS (166 PROVED EQUIVALENT): THE RULES THAT PRUNE CALLS WITHOUT LOSING A PROVED CONSTRUCT, AND THE LARGEST RULE OVERALL.

Rule (kind/feature=bin)	Calls pruned	Fraction	Equivalent lost
adder/out_reads=no	14	0.067	0
subtractor/width=2	8	0.038	0
subtractor/out_reads=no	4	0.019	0
adder/npn_sum=low	3	0.014	0
subtractor/npn_carry=all	3	0.014	0
unrestricted best	79	0.376	79

record, hence the operator F1, unchanged (0.0000). By the rule H7 is supported by the letter of the rule (weak evidence). The tie the rule was written for, the calibrated flow against the L0-stop netlist (both add nothing to the text), does not separate: their operator type-F1 is 0 against 0. The verdict rests on the calibrated flow against the uncalibrated one, where the calibrated side recovers no operator at all, and on the flatten clause, which holds by construction. Two cautions: the separating level is small (the uncalibrated detectors count mostly wrong constructs), and the flatten comparison holds the record fixed by construction, because the operator F1 reads the recovery record and AST and CFG read the text. The operator F1 does not replace them; it is the measure that sees recovery, where they see only form.

RQ-L2: Loki against the language model on the same designs. Round 1 scored the language model on Q1 and Q2 of VerilogEval, and Loki's AST and CFG only on micro designs. We ran `loki re` on all 150 VerilogEval netlists at f2 and f3, calibrated (a calibration fitted on the 117 train designs of VerilogEval itself) and not, and scored the output Verilog (`design.v`) against the reference `.sv` with the round-1 AST and CFG code. 21 references that pyverilog cannot parse have no AST value for any candidate, and 6 f3 outputs exceed the CFG probe's size limit (`l2_coverage.csv`). On the designs both are scored on (Table XVI), Loki's f3 output has AST similarity 0.763 in Q1 and 0.460 in Q2 (common designs, $n = 36$ and 36) against the model's 0.894 and 0.761, and CFG similarity 0.751 and 0.743 against 0.942 and 0.890; every gap is significant by a paired Wilcoxon test (`l2_paired.csv`). The calibration was fitted on the train designs and these means cover all 150, so we also give the held-out test split alone (`l2_testsplit.csv`): on its 16 Q1 and Q2 designs both tools score on, AST is 0.705 for Loki against 0.893 for the model and CFG 0.802 against 0.927, the same direction. (The calibrated f3 flow commits in only 7 designs, so the fit can change few outputs.) Across the four quartiles (Table XVII) the f3 AST over all scored designs ($n = 38$, 38, 32, 21; a different, larger design set than the common rows above) reads 0.754, 0.455, 0.605 and 0.505, so it falls from Q1 to Q4 without being monotone. The 21 references that do not parse fall in Q1 to Q4 as 0, 1, 5 and 15, so Q3 and Q4 are scored on the designs that parse, and the apparent fall may partly reflect which designs those are. The CFG stays flat (0.750 to 0.768). The calibrated f3 flow commits almost nothing, so its output is the L0 netlist; the uncalibrated flow, which does commit, scores lower, not higher.

What the correctness column proves is narrow. Every one of the 150 runs of each set ends `final_equiv = equivalent` (150 for the calibrated f3 set), which says that the flattened output is formally equal to the *input netlist*. It does not say the output matches the reference RTL: that netlist itself passes the official testbench in only 128 of the 150 designs (the round-1 self-check; inferred latches and asynchronous resets are the failures), so Loki's output is exactly as correct as its input, no more. The model is checked against the testbench directly, but only its correct samples enter the similarity columns. On operator recovery (RQ-L7) the model is far ahead: on the common Q1 and Q2 designs its operator type-F1 is 0.465 (functionally correct samples elaborated with the same Odin III front as the key) against 0.081 for Loki's calibrated f2 run and 0.074 uncalibrated (the calibrated f3 run recovers no operator, 0.0, so the comparison uses f2, a different floor from the one AST and CFG use). By the rule, H2 is inconclusive: Loki is below the model on

both metrics in both quartiles (yes), the equivalence holds in all runs (yes), and the similarity does not stay flat from Q1 to Q4 (no). The positioning this measures is that the model is readable and recovers the operators of small designs but is unverified, and Loki is verified but partial; neither number alone is the result, and the arm that combines them (a model given the ReverseProvenance record, L8) was not run.

RQ-L4: which evidence family drives the posterior? The calibrated flow scores every hypothesis from a per-detector prior plus feature log-likelihood ratios in four families (`npn`, `wl`, `l2`, `struct`). `loki re` produces no `l2` feature (those exist only in the ReverseProvenance document), so that family is not applicable. For each remaining family we refit the calibration on the train designs of the micro and VerilogEval f2 floors with the family removed and its weight set to 0, and ran the 49 held-out designs (Table XVIII; 1,558 ground-truth instances). Held-out micro-F1 is 0.0159 with the defaults, 0.0038 with the full calibration and 0.0051 with priors only. The full-minus-priors gap is -0.0013, below the 0.02 floor of the rule, so H4 is inconclusive: no family carries measurable performance, because the features add nothing detectable over the per-detector priors. One held-out design (`ve_Prob153_gshare`) holds most of the ground truth and nothing in it is matched; with it left out (257 instances) the micro-F1 is 0.0553 with the defaults, 0.0225 full and 0.0295 priors only (gap -0.0070), so the verdict does not change; on that subset the defaults score higher than either calibrated row. The fit says so itself: the full calibration sets the `npn` weight to 0 and shrinks `struct` and `wl` (`l4_weights.csv`). What calibration does is prune, from the many counted constructs of the defaults to a handful, and rank: the AUC of p_{comp} over the live hypotheses rises from 0.437 to 0.885 (over different pools, since calibration rejects more).

RQ-L5: is the must/may bracket an uncertainty measure? Following the graph-sandwich framing (must is what L1 proved and committed, may adds the candidates it lists), the bracket of a run is read from `reverse-provenance.json` and its width is the candidate fraction of the cells. H5a: the width correlates negatively with the design's F1. On the uncalibrated VerilogEval f2 runs (90 designs with ground truth) the Spearman correlation is 0.087 ($p = 0.417$), and -0.346 on the calibrated run that lists candidates from $p = 0.1$ (Table XIX), so H5a is not supported. The committed fraction does correlate with F1 (0.485), but trivially: committed cells are the matches. As a soundness check of the bracket itself, 0.840 of the committed cells lie inside the scored ground-truth footprints (must precision), the committed cells cover 0.240 of the truth, and the may bracket contains 0.625 of it. H5b: of 210 `cone_check` (SAT) calls across all round-2 runs, 166 end `equivalent`; the largest structural rule that loses no such verdict (adder/out_reads=no) prunes 0.067 of the calls, against the 0.3 the rule asked for (Table XX), so H5b is not supported. The prover budget is spent confirming true hypotheses, and the structural features the calls already carry do not separate the rest; the truth-table checks (2,876) are cheap already. The rule search covers single-bin rules (one feature and one bin of one kind) on the same data, so 0.067 is an upper bound for such rules, not a measured saving; the calls repeat across the eight run sets. The bound that holds for any rule is simpler: a rule that loses no `equivalent` verdict can prune at most the 44 calls that were not (0.21 of all).

TABLE XXI
RQ-L9, PRIMARY SET P (M2N+V2N, F2, UNCALIBRATED): COUNTED CANDIDATES, FALSE ONES, AND THE SHARE OF THE FALSE ONES PER CLASS (RULES OF `EXPERIMENT.MD`).

kind	cand.	false	same ext.	frag. rel.	frag. other	straddle	partial	noise
ff_en	1,070	1,067	0.01	0.99	0.00	0.00	0.00	0.00
mux_bit	344	336	0.00	0.61	0.39	0.00	0.00	0.00
eq_const	1,105	1,096	0.00	0.18	0.82	0.00	0.00	0.00
register	155	116	0.03	0.64	0.15	0.19	0.00	0.00

TABLE XXII
RQ-L9 (H9B), SET P: GROUND-TRUTH INSTANCES WITH AT LEAST TWO CELLS, THE SHARE WHOSE FRAGMENT UNION COVERS AT LEAST 0.8 OF THEM (q), THE SAME WITH CANDIDATES OF ANY KIND, AND THE MEAN FRAGMENT COVERAGE.

kind	GT	matched	q frag.	q any	mean cov.
ff_en	156	3	0.122	0.160	0.618
mux_bit	368	2	0.005	0.136	0.016
eq_const	995	5	0.040	0.112	0.089
register	289	39	0.194	0.228	0.497

G. Round 3: fragments or noise, and where the words are

Round 2 left a puzzle. Uncalibrated, the detectors propose many candidates that almost never match (Table XXI); calibrated, they commit almost nothing. Peter's question (Appendix A) is why netlists are hard to bring back to HDL when the structure is there to see: muxes, wide signals that behave as words, state feedback with decode and select, and input and output words that mark where the datapath lies. Round 3 is compute-only on the round-2 runs and the L0 ground truth (no language-model call, no new detector). Unlike round 2, its rules were committed before any number was computed (commit `5f8de24`, `experiments/round3/EXPERIMENT.md`); every figure below is a macro generated from the summary CSVs of `experiments/round3`. We added the runs the round-2 sets lacked (micro f3, calibrated and not, and micro f1-k6, uncalibrated; one micro f3 design timed out) and score every set, but the decisions use the primary set P: the micro and VerilogEval f2 runs without calibration.

RQ-L9: are the wrong candidates fragments of real words? A false candidate is a counted candidate the one-to-one scorer does not match. We classify each (rules in the protocol): the same extent as a ground-truth instance of its family (a lost tie or a type slip), a fragment of a larger instance of its family, a fragment of an instance of another type, a straddle of several, a partial overlap, or noise. Over the 2,615 false candidates of the four kinds in P, 0.583 are fragments of an instance of the matching or a related family, and 0.000

TABLE XXIII
RQ-L9 FOR EVERY RUN SET (FOUR DECISIVE KINDS POOLED): RUNS, CANDIDATES, FALSE CANDIDATES AND THE SHARE THAT ARE RELATED FRAGMENTS, OTHER-TYPE FRAGMENTS, NOISE.

set	runs	cand.	false	frag. rel.	frag. other	noise
M2N	65	705	686	0.28	0.69	0.00
M2C	65	71	65	0.95	0.05	0.00
V2N	150	1,969	1,929	0.69	0.30	0.00
V2C	not run					
V3N	150	5,005	4,997	0.29	0.71	0.00
V3C	150	2	0	0.00	0.00	0.00
M3N	64	1,406	1,405	0.23	0.77	0.00
M3C	not run					
M1N	68	496	469	0.38	0.57	0.00

TABLE XXIV
RQ-L10, F2 FLOOR, MICRO AND VERILOGEVAL POOLED, RELAXED RULE ($\tau = 0.75$): THE SHARE OF GROUND-TRUTH WORD CONSTRUCTS SEEDED BY A PROPAGATED GROUP WITHIN h HOPS OF AN I/O PORT WORD, AND THE SHARE NEVER REACHED.

kind	class	GT	$h \leq 1$	$h \leq 2$	$h \leq 3$	$h \leq 4$	never
register	datapath	289	0.57	0.66	0.66	0.69	0.31
mux	datapath	90	0.40	0.41	0.41	0.41	0.59
mux_tree	datapath	13	0.23	0.23	0.23	0.23	0.77
adder	datapath	2,124	0.00	0.00	0.00	0.00	1.00
subtractor	datapath	20	0.20	0.20	0.20	0.20	0.80
comparator	datapath	23	0.30	0.30	0.30	0.30	0.70
counter	datapath	23	0.00	0.00	0.00	0.00	1.00
eq_const	control	990	0.61	0.61	0.61	0.61	0.39
fsm	control	16	0.06	0.06	0.06	0.06	0.94

are noise: every false candidate lies inside some ground-truth construct. Per kind the related-fragment share is 0.988 for `ff_en`, 0.638 for `register`, 0.610 for `mux_bit` and only 0.175 for `eq_const`; 3 of four kinds reach 0.5, so H9a is supported. The split is informative. The `ff_en` candidates are two-cell pieces (an enable multiplexer and its flip-flop) of a larger construct, most of them in one VerilogEval design (`ve_Prob153_gshare`), where they sit inside a mux tree; the word-level ground truth cannot match a piece, a grouping problem. The `eq_const` candidates are 0.822 fragments of an instance of another type (mostly single-LUT `$and` and `$or` words, registers and `fsm` instances), a detector problem: the LUT computes something that looks like a constant compare and is not one. The `mux_bit` candidates are 0.390 inside an instance of another type (`$shiftx` words among them). H9b asks whether grouping the fragments would recover the word: the share of ground-truth instances whose fragment union covers 0.8 of them is 0.065 pooled (`ff_en` 0.122, `register` 0.194, `eq_const` 0.040, `mux_bit` 0.005), and no kind reaches 0.5, so H9b is not supported. Even with candidates of any kind the cover stays low (`register` 0.228, `mux_bit` 0.136). The fragments exist, but they do not tile the words: a merge step over the present detector output would recover few words. The result is sensitive to the largest design: with `ve_Prob153_gshare` left out (an exploratory check, not the rule) 1,863 false candidates remain, 0.415 of them related fragments and 0.563 fragments of another type, so the pooled share falls under the 0.5 of the rule and H9a would be inconclusive; `ff_en` stays at 0.966 and `register` drops to 0.333. The other sets (Table XXIII) have fewer candidates when calibrated, and the gate floor f3 proposes far more false `eq_const` candidates than any f2 set.

TABLE XXV
RQ-L10 BY DATASET AND FLOOR: DESIGNS, DESIGNS WITH A PORT WORD, INPUT AND OUTPUT PORT WORDS, AND THE DATAPATH AND CONTROL SHARES REACHED WITHIN TWO HOPS.

dataset	floor	designs	with port word	in words	out words	datapath $h \leq 2$	control $h \leq 2$
micro	f2	68	54	158	22	0.35	0.68
micro	f3	65	51	150	29	0.63	0.99
micro	f1-k6	68	54	158	38	0.66	0.78
veval	f2	150	63	103	13	0.07	0.60
veval	f3	150	63	103	13	0.94	0.88

TABLE XXVI
RQ-L11, GROUND TRUTH: FEEDBACK REGISTERS, CONSTANT-COMPARE DECODERS ON THEM, SINKS (MUX SELECTS AND ENABLES) AND CHAINS, AND THE MUXES WITH A SELECT (ENABLED WORDS) WHOSE SELECT (ENABLE) IS A CHAIN SINK.

dataset	floor	designs	with fb.	with chain	fb. reg.	fsm flag	decoders	sinks	chains	mux S chain/all	enable chain/all
micro	f2	68	3	1	3	0	11	3	12	3/41	0/3
micro	f3	65	3	1	3	0	11	3	12	3/41	0/3
micro	f1-k6	68	3	1	3	0	11	3	12	3/41	0/3
veval	f2	150	55	12	197	16	41	30	89	15/43	15/152
veval	f3	150	55	12	197	16	41	30	89	15/43	15/152

RQ-L10: do the I/O words find the datapath? A top-level port bus is a word the reverse engineer usually has by name. A group of cells is a propagated word if, bit by bit, a known word drives (or is driven by) one cell per bit of the group with the same cell type, truth table and pin, and at least three quarters of the bits qualify (the strict rule asks for all); we propagate for up to four hops. A ground-truth construct is reached when a group of at most h hops lies for the most part inside it. In the f2 floor, 2,582 datapath word constructs (registers, muxes, adders, comparators, counters) and 1,006 control constructs (`eq_const` and FSMs) have two or more bits; 0.095 of the datapath and 0.605 of the control constructs are reached within two hops, so H10a is not supported and H10b is not supported. Both numbers need reading. The datapath share is pooled and one VerilogEval design alone holds most of the 2,124 adder constructs, which the rule never reaches because the carry chain breaks the bit-for-bit isomorphism; the mean of the per-design shares is 0.326 (datapath, 113 designs) against 0.286 (control, 51 designs), a difference of the sign the hypothesis expects but not of its size. Registers are reached best (0.574 at one hop, 0.664 at two), muxes less, adders and counters almost never, and a second hop adds little: bit-sliced isomorphism stalls after the first layer of logic. The control constructs are reached not because FSM logic follows the ports but because the constant compares read input ports directly (`eq_const` 0.614 at two hops); the FSMs themselves are not. The strict rule reaches 0.032 and 0.070, and the gate floor f3 0.914 and 0.894: at f3 a cell group lands inside a gate-level footprint almost by construction, so that figure measures the floor, not the idea. The honest summary is that I/O words seed registers and input-side muxes well, and nothing else.

RQ-L11: the FSM chain. Measurement only. In the f2

TABLE XXVII
RQ-L11, DETECTORS: CHAIN ELEMENTS PROPOSED BY A COUNTED CANDIDATE WITH JACCARD AT LEAST 0.5 OF ANY KIND (IN BRACKETS: OF THE RIGHT SCORED TYPE), CHAINS WITH ALL THREE ELEMENTS PROPOSED, AND GROUND-TRUTH FSM INSTANCES MATCHED BY AN `FSM` CANDIDATE.

set	fb. register	decoder	sink	complete chains	fsm
M2N	0/3 (0)	1/11 (0)	1/3 (0)	0/12 (0)	0/0
M2C	0/3 (0)	0/11 (0)	0/3 (0)	0/12 (0)	0/0
V2N	41/197 (26)	15/41 (9)	5/30 (0)	5/89 (0)	6/16
V2C	3/197 (0)	1/41 (0)	2/30 (0)	1/89 (0)	0/16
V3N	13/197 (3)	6/41 (2)	3/30 (0)	4/89 (0)	3/16
V3C	4/197 (0)	2/41 (0)	2/30 (0)	2/89 (0)	1/16
M3N	0/3 (0)	0/11 (0)	0/3 (0)	0/12 (0)	0/0
M3C	0/3 (0)	0/11 (0)	0/3 (0)	0/12 (0)	0/0
M1N	0/3 (0)	0/11 (0)	0/3 (0)	0/12 (0)	0/0

ground truth of VerilogEval (150 designs) 197 registers feed back to themselves, of which 16 the front end flags as FSM state; 41 constant-compare decoders read such a register, and the chains (register, decoder, select or enable) occur in 12 designs. Of the muxes with a select, 15 of 43 (VerilogEval) and 3 of 41 (micro) take it from such a decoder, 0.214 together, so the FSM, decode, select chain is real but covers a minority of the control muxes of these small designs. The detectors see little of it (Table XXVII): in V2N they propose 41/197 feedback registers, 15/41 decoders, 5/30 sinks, all three elements of 5/89 chains, and 6/16 FSM instances as an `fsm` candidate.

Implication for a word-first detector. The wrong candidates are not noise: none lies outside the ground-truth constructs, so the evidence is there. The fragment share depends on the kind and, for the pooled figure, on one design: `ff_en`, `mux_bit` and (less so) `register` fail as pieces of larger constructs, while `eq_const` fails as a detector problem (the local pattern is too weak, and a word-level view, the select and enable fan-out across a word, must decide whether a compare is a decoder). Because the fragments do not tile the words, grouping the present candidates is not enough, and a detector should construct the word first and score it as a unit. Concretely, we propose to find (i) register words by shared clock, enable and reset across flip-flops; (ii) datapath words by per-bit cones that are NPN-isomorphic, seeded from the ports but not limited to them (the port words reach registers and nothing past the first layer); (iii) control nets by select and enable fan-out across a word; and (iv) FSMs by a feedback strongly connected component plus the decoders on it. Related work attacks the word problem along the same lines: WordRev [6] derives word-level structure from a gate netlist by matching sequences of word-level operations and by propagating from known words, and DANA [7] recovers registers from the dataflow between flip-flops. We have not run either tool on these sets and do not claim a comparison. The next round builds this detector (L8, the language model given the ReverseProvenance record, stays deferred).

VI. DISCUSSION

Three observations follow. First, *the answer depends on what the flow erased*. The structures that survived synthesis recover almost perfectly without any intelligence beyond reading the pass's own facts; the structures synthesis erased (the counter behind its flip-flop, the mux behind its gates) recover weakly, and no amount of honest Bayesian machinery changes that without better detectors or a model. Second, *the graph features carry the recovery*: the GNN only beats 1-WL when node, graphlet, WL and NPN features combine, which is the same lesson as the clustering table (community methods find the hierarchy, centroid methods do not). Third, *the benchmarks made the limits visible*: the renamed-instance protocol proved naming neutrality over 105,646 identifiers, and the same protocol exposed a rename-script defect (library `.blackbox` model names were renamed, which silently removed every hard proposal on renamed floors) and the spectral determinism defect – both fixed or reported, and both invisible without the protocol.

A fourth observation comes from round 2 (Section V-F): *readable and verified are different properties, and the two tools hold one each*. The language model produces RTL whose similarity to the reference is high and whose operators it names (operator type-F1 0.465 on Q1 and Q2, correct samples elaborated to RTL), but nothing proves the text equals the netlist; Loki's output is formally equal to its input in all 150 runs and recovers few operators (0.081, calibrated f2 run; the f3 run recovers none), and its similarity to the reference is lower in every cell. The f3 failure of round 1 may be the detectors' vocabulary and not missing information (a LUT6 re-map reaches the f2 level in-sample but not on the held-out designs, 0.87 against 0.51, so the question stays open), the three refinements we expected to help (evidence-family weighting, the width of the must/may bracket as an uncertainty measure, a structural prefilter before the prover) did not on the registered tests, and the one new instrument that earned its place is the operator F1, which sees recovery where AST and CFG see only form.

A fifth observation comes from round 3 (Section V-G): *the wrong candidates are pieces of real constructs, not noise, but the pieces do not tile the words*. Every false candidate of the four kinds lies inside some ground-truth construct, about half as a fragment of an instance of a related kind (H9a supported, sensitive to one design), yet a word's fragments rarely cover it (H9b not supported). Propagation from the I/O words reaches registers but not the arithmetic or the FSMs (H10a not supported, H10b not supported), and the FSM, decode, select chain covers 0.214 of the control muxes. The next detector is therefore word-first, not a regrouping of the present one.

VII. CONCLUSION

Loki III asks how far netlist-to-RTL reverse engineering gets without a language model. On the benchmarks of this paper the answer is: the already-reverse-engineered structures, perfectly; the rest, weakly but measurably, with calibration

that reports its own error, a graph stage whose features beat 1-WL, and clusters that carry the designer's hierarchy. The methodology – provenance-tracked labelled datasets plus the renamed-instance protocol – is the part we would export to any future comparison, with or without a model.

APPENDIX A
RESEARCH LOG

This appendix records the order in which the questions were asked, what each result suggested, and what became of the next set of hypotheses, so that the progress of the work is visible and not only its end state. The decision rules of round 2 are in `experiments/round2/EXPERIMENT.md`; their thresholds were set by the implementer during the round (the file was committed after the scripts and the result CSVs), so they are not pre-registered. The rules of round 3 (`experiments/round3/EXPERIMENT.md`) were committed before any round-3 number was computed (commit `5f8de24`).

Round 0 (2026-10-21 to 22): build. Question: how much structure does Loki recover without a language model? Built L0 (provenance-labelled, anonymised datasets and the renamed-instance neutrality check), L1 (Bayesian hypotheses with calibration and proofs) and L2 (graph analysis).

Round 1 (2026-10-23): first results. Section V-B to Section V-E. Hard blocks are fully recovered, because synthesis had already reverse engineered them; soft structures recover weakly; on the cmos6 gate netlist (f3) L1 recovers nothing and AST and CFG do not move; the GNN beats 1-WL; and the language model is strong on the simple VerilogEval quartiles but its output is unverified.

What round 1 suggested. (i) The f3 zero beside a weak f2 result: is the failure the detectors' LUT vocabulary, not missing information? (ii) The language-model and Loki figures sat on different benchmarks: put them on the same 150 designs. Our working position is that the model is readable but unverified on small designs, Loki is verified but partial, and the combination is the contribution. (iii) The calibrated flow commits little: which evidence family earns that? (iv) The reverse-engineering result is a smaller graph bracketed by what is proven (must) and what is possible (may), in the style of the graph-sandwich problem: is the width of that bracket an uncertainty measure, and does a cheap structural test before the prover keep recall? (v) AST and CFG tie where the recovery differs: is there a metric that sees it?

Round 2 (2026-10-24): the hypotheses and what happened. Table XXVIII. The operator F1 is the one hypothesis the rule supports, by its letter and on weak evidence (H7). The LUT6 re-map reaches the f2 level in-sample (0.87) but not on the held-out split (0.51, too few matches for the rule), so H1 is inconclusive. The head-to-head puts Loki below the model on every cell with its output equal to its input in all runs, but the similarity does not stay flat with complexity (H2, inconclusive). The evidence families add nothing over the priors on held-out designs, because the calibration already zeroes or shrinks them (H4, inconclusive). The bracket width is not an uncertainty measure on the registered set (H5a, not supported; an exploratory look at calibrated runs that list candidates from $p = 0.1$ gives Spearman -0.346), and a structural prefilter cannot cut the prover without losing proved constructs (H5b, not supported). The head-to-head (Section V-F) fills in the positioning of the discussion.

What round 2 suggested. The calibrated flow commits almost nothing and the uncalibrated one proposes many wrong candidates. Are the wrong candidates fragments of real word constructs (a grouping problem) or unrelated (a detector problem)? Peter (2026-10-24) put the same question from the reverse-engineering side: "for Loki - the question is from a reverse engineering perspective why is it hard to reconstruct to HDL - going from if structures (which we know are muxes) and then down it seems like many of the patterns are constant - so hypothesizing that things are a bus (and by a bus I mean a signal that could be a traditional bus, but also a wide wordsize signal that has regular oprtations to it - for example figuing out that there is a word in the datapath or control then allows us to look for patterns to reconstruct. Control flow is usually in the form of muxin and FSMs have pattern of feedback (state equals next state) and then control based on state. How do we find these patterns better?" and "Also, the outputs (words) and the inputs as words also provide hints of where the words are".

Round 3 (2026-10-24): the hypotheses and what happened. Table XXIX; Section V-G. Rules committed first (`5f8de24`). H9a (wrong candidates are related fragments) is supported with related-fragment share 0.583 of 2,615 false candidates and no noise, but not robust to leaving out the largest design (0.415); H9b (fragments cover the word) is not supported at 0.065; H10a (datapath words reached from the ports within two hops) is not supported at 0.095 and H10b (control is not) not supported at 0.605; the FSM chain (L11) is measured only. The next step is a word-first detector (register words by shared clock, enable and reset; datapath words by per-bit isomorphic cones; control nets by select fan-out; FSMs by feedback components and their decoders); L8 stays deferred.

Next. L3, a cross-suite GNN (train on one suite, test on the other); L6, a motif null model for the graphlet counts; and L8, a language model given the ReverseProvenance document (deferred by Peter), which is the experiment that tests the combination directly.

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TABLE XXVIII

RESEARCH LOG: THE ROUND-2 QUESTIONS, THEIR STATUS AND THE DECIDING NUMBER (ALL GENERATED FROM `EXPERIMENTS/ROUND2`).

RQ	Hypothesis	Round	Status	Deciding number
L1	f3 failure is detector vocabulary	2	inconclusive	micro-F1 LUT6/f2 = 0.87 (all, in-sample), 0.51 (test)
L2	Loki below the LLM on AST/CFG, flat over Q1–Q4	2	inconclusive	AST Q1/Q2 0.763/0.460 vs 0.894/0.761
L4	one or two evidence families carry the posterior	2	inconclusive	full minus priors-only micro-F1 -0.0013
L5a	bracket width falls with per-design F1	2	not supported	Spearman 0.087 ($p = 0.417$)
L5b	structural prefilter cuts prover calls, keeps recall	2	not supported	best zero-loss rule prunes 0.067
L7	operator F1 separates AST/CFG ties	2	supported by the letter of the rule (weak evidence)	tie pair dF1 0.0173; flatten dAST 0.0422
L3	cross-suite GNN	next	not run	–
L6	motif null model	next	not run	–
L8	LLM with reverse provenance	next	deferred (Peter)	–

TABLE XXIX

RESEARCH LOG: THE ROUND-3 QUESTIONS, THEIR STATUS AND THE DECIDING NUMBER (ALL GENERATED FROM `EXPERIMENTS/ROUND3`).

RQ	Hypothesis	Round	Status	Deciding number
L9a	wrong candidates are fragments of GT words	3	supported	related-fragment share 0.583 of 2,615 false (noise 0.000)
L9b	fragments of a word cover it	3	not supported	share with coverage ≥ 0.8: 0.065
L10a	datapath words reached from I/O within 2 hops	3	not supported	share reached 0.095 (per-design mean 0.326)
L10b	control words are not	3	not supported	share reached 0.605 (per-design mean 0.286)
L11	FSM chain: feedback, decode, select	3	measurement only	0.214 of muxes with a select are chain sinks

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